COMMUNITY PARKING MANAGEMENT

User Manual

Guide for Administration, Control and Owners

> **Purpose of this manual**
> Explain the current operational use of the application: access, dwellings, vehicles, slots, QR codes, access control, incidents, requests, audit, settings and printing.

ADMIN	CONTROL	OWNER

Updated: November 2026

Platform: GitHub Pages + Supabase

1. Introduction

The application centralises community parking access control through dwellings, vehicles, slots, QR codes, registration plates, incidents and requests. It can be used on desktop and mobile devices.

> **Main idea**
> A vehicle can be registered without having parking access. Access depends on its status, slot assignment when applicable and QR status. Motorcycles are controlled by registration plate and do not consume a slot.

1.1 User roles

Role	Main use	Main screens
ADMIN	Full management and supervision	Summary, Data Review, Users, Dwellings, Vehicles, Incidents, Requests, Audit and Settings.
CONTROL	Access validation and incident registration	Control and Incidents.
OWNER	Manage own dwellings and vehicles	Home, Dwellings, Vehicles, My Incidents and My Requests.

If a person has more than one role, they can switch profile from the role selector. ADMIN opens Summary; CONTROL opens Control; OWNER opens Home.

1.2 What this guide covers

- Access, password recovery, ES/EN selector and mobile navigation.
- Vehicle status, slots, QR codes and temporary plates.
- ADMIN, CONTROL and OWNER functions.
- Incidents, follow-ups and requests.
- Audit, data review and settings.
- Individual and bulk QR printing.

2. Access and navigation

2.1 Sign-in and password

1. Open the application address in your browser.
2. Select ES or EN if you want to change the interface language.
3. Enter your email and password.
4. If you have multiple roles, choose the profile you want to use.

> **Password recovery**
> Use “Forgot your password?” and always use the most recent link received by email. Older reset links may become invalid. The password must meet the rules shown by the application.

2.2 Language

The selected language is stored in the browser. If no preference is saved, the application attempts to use the browser language. Real data such as names, plates or user-entered values is not automatically translated.

2.3 Desktop and mobile navigation

- On desktop, the main sections appear in the top navigation bar.
- Search finds vehicles, temporary plates, incidents and, depending on role, dwellings and requests.
- On mobile, use the hamburger menu and bottom navigation.
- Audit is intended for desktop use.
- Summary/Home cards open destination lists with the relevant filter already applied.

2.4 My data

Open My data from the user name or mobile menu. Editable information depends on role and permissions.

3. Core concepts and rules

3.1 Vehicle status

Status	Meaning	Slot	QR	Access
Active	Vehicle authorised to use the parking area.	Yes for cars; not for motorcycles.	Active or Revoked.	Yes, unless QR revoked.
Unassigned	Registered but without a slot.	No	No QR	No
Suspended	Temporarily blocked.	No	No QR	No
Deactivated	Administratively deactivated.	No	No QR	No

> **Consistency rule**
> An Active car must have a slot. An Unassigned, Suspended or Deactivated vehicle does not keep an operational slot or QR. Active motorcycles do not use slots or QR codes.

3.2 Slots

- Each dwelling has a maximum of 2 slots for vehicles that consume a space.
- More vehicles can be registered than there are available slots.
- Assigning a slot to an Unassigned car makes it Active and creates a QR credential.
- Changing slot for an already Active car keeps its authorisation/QR.
- Releasing the slot makes the vehicle Unassigned and removes its operational QR.
- Motorcycles do not consume slots.

3.3 QR and temporary plates

The QR identifies the vehicle authorisation. A valid temporary plate does not create a different QR: the same QR remains in use and Control validates either the habitual plate or the current temporary plate. Revoking invalidates that credential; renewing creates a new one.

4. ADMIN profile

4.1 Summary and Data Review

Summary shows operational indicators and items requiring attention. Navigation cards clear previous filters and apply the relevant filter in the destination list.

- Active, unassigned, suspended and deactivated vehicles.
- Open requests and incidents.
- Current or expiring temporary periods.
- Consistency alerts across dwelling, slot and QR data.

Data Review detects issues such as duplicate plates, duplicate slots, active vehicles with inconsistent authorisation, motorcycles with slot/QR data or incompatible temporary periods.

4.2 Users, Dwellings and Vehicles

Users manages profiles and access. Dwellings shows responsible owner, status and occupancy. Vehicles supports search, filters and state-dependent actions.

- Edit permitted data.
- Assign, change or release a slot.
- Manage temporary plate periods.
- View, revoke or renew QR when applicable.
- Replace an active vehicle.
- Suspend or deactivate.
- Review requests, incidents and history.

4.3 QR printing

> **Same QR in every flow**
> Individual and bulk printing use the same canonical QR image linked to the vehicle. Printing never creates a new credential.

- Individual print: habitual plate + QR.
- Bulk print: select vehicles and press “Print QR”.
- Motorcycles, vehicles without a slot or without an available active QR are skipped.
- Bulk printing places 2 cards on each A4 sheet.
- QR physical size is the same in individual and bulk printing.
- For the cleanest result, print at 100% and disable browser headers/footers.

5. Incidents

> **Usage rule**
> Incidents are registered from the CONTROL flow. ADMIN reviews, follows up and resolves them; there is no need for a separate administrative incident-creation flow.

5.1 Main incident types

Type	Typical situations
Unregistered plate	The observed plate does not exist in the system.
Unauthorised access	Suspended, deactivated, unassigned, revoked QR or another unauthorised condition.
QR does not match plate	The scanned QR belongs to a different vehicle.
Misuse	QR lending or use of another vehicle's QR.
Immobilised / possible abandonment	Long-term immobilisation, possible abandonment, inspection/repair issue.
Blocking / wrong space / common area	Blocks routes, wrong bay, common area or another parking-location issue.
Other	Exceptional case not covered above.

5.2 Registering from Control

5. Scan a QR or search for a registration plate.
6. Review the authorisation result.
7. Press Register incident.
8. Choose the type and observed situation.
9. Add a clear note and the required photo/evidence.
10. Confirm the incident.

The application prevents equivalent duplicates on the same day and may direct the user to add a follow-up when an existing case already covers the situation.

6. Incident follow-up and resolution

6.1 Follow-ups

- At most one follow-up per incident per day.
- Photo/evidence is part of the follow-up when applicable.
- History retains who recorded each action and when.

6.2 Linked incidents for QR misuse

Scenario	Expected result
QR A used by vehicle B belonging to the same owner	“QR does not match plate” incident on the observed vehicle.
QR A used by B belonging to another owner	Linked incidents for QR lending/misuse; if B is not authorised, unauthorised access is also recorded.
QR A + unknown plate	QR misuse/lending incident plus unregistered plate.

6.3 Resolution

ADMIN reviews the case, follow-ups, vehicle and evidence. Resolution records the applied action and closing comment. The owner can see the resolution in history when applicable.

> **Traceability**
> Do not delete resolved incidents or requests simply to clean up the interface. History is part of operational control and auditability.

7. CONTROL profile

7.1 QR lookup

11. Open Control.
12. Start the scanner and allow camera access when requested by the browser.
13. Scan the vehicle QR.
14. Check habitual plate, current temporary plate if any, vehicle status and authorisation.

A revoked QR remains recognisable as an invalidated credential, allowing Control to identify the vehicle and record the appropriate incident.

7.2 Plate lookup

Manual plate search can validate a vehicle when no QR is presented. If a temporary plate is currently valid, the application links it to the habitual vehicle.

7.3 Authorisation result

Result	Action
Authorised	Allow access if the observed plate matches the habitual or current temporary plate.
Unauthorised	Review the reason and register an incident when appropriate.
Not registered	Register an unregistered-plate incident when applicable.
QR/plate mismatch	Use the QR misuse / mismatch flow.

7.4 Manual registration

When the event does not originate from a QR scan or a specific plate search, CONTROL can start a manual incident from the general Control screen.

8. Requests

Requests allow owners to ask for changes that require administrative review. The initial status is Open; ADMIN can approve or reject and records the outcome.

Type	When used	Effect when approved
Activation	Suspended or Deactivated vehicle.	Car: becomes Unassigned without slot/QR until reassigned. Motorcycle: becomes Active.
QR renewal	Active car with slot and Revoked QR.	Creates a new Active QR credential.
Long stay	Travel, immobilisation or another authorised prolonged stay.	CONTROL sees the authorisation during the approved period.

> **Activation is not immediate access**
> When activation of a Suspended/Deactivated car is approved, it becomes Unassigned. Access returns only after a slot is assigned.

8.1 Statuses

Status	Meaning
Open	Pending review.
Approved	Accepted and the corresponding effect applied.
Rejected	Declined without applying the requested change.

9. OWNER profile

9.1 Home and My dwellings

Home summarises dwellings, vehicles, slots, motorcycles, requests and incidents. Summary cards open filtered lists. My dwellings shows status, occupancy and associated vehicles.

9.2 My vehicles

Action	When shown / what it does
Edit	Updates editable vehicle data.
Temporary plates	Manages temporary registration periods.
QR	Available for Active car with slot and Active QR.
Print	Prints habitual plate + QR using the same canonical image as ADMIN.
Assign/Change/Release slot	Available according to vehicle status and authorisation.
Replace	Replaces an active vehicle according to authorisation rules.
Long-stay notice	Creates a long-stay request.
Request QR renewal	Available with Revoked QR.
Deactivate	Moves the vehicle to Deactivated.
Request activation	Available for Suspended or Deactivated vehicles.

> **Temporary plate and QR**
> A current temporary plate does not change the QR. CONTROL validates the vehicle credential and accepts the temporary plate during the authorised period.

10. My incidents and My requests

10.1 My incidents

Owners can review incidents linked to their vehicles, including type, dates, follow-ups and resolution. Owners cannot resolve incidents from this screen.

10.2 My requests

Shows Open, Approved and Rejected requests. Administrative resolution and comments remain visible in history.

10.3 Good owner practices

- Keep the habitual plate up to date.
- Create a temporary period instead of replacing the habitual plate for temporary changes.
- Do not share the QR with another vehicle.
- Request QR renewal only when the QR is Revoked.
- Use Replace for permanent vehicle changes.

> **Printed QR**
> A printed QR remains valid while the vehicle credential remains Active. If it is revoked and renewed, previous printed copies no longer authorise access.

11. Audit, Settings and data

11.1 Audit

Audit helps reconstruct relevant administration and control actions. It is intended primarily for administrative desktop use.

11.2 Settings

Settings groups operational parameters available to ADMIN, including permitted data/import tools. Options shown depend on user permissions.

11.3 Dwelling and owner import

Bulk import is performed from Settings > Data. Review the preview, records to create/update and any matching issues before applying changes.

> **Important**
> Do not edit the database directly to solve an operational issue when the application provides an equivalent action. Use Data Review, administrative screens and Audit to preserve traceability.

11.4 Data review

- Review inconsistencies before applying a correction.
- Confirm dwelling, slot, vehicle status and QR.
- Check current temporary periods and open requests.
- Run the review again after a significant correction.

12. Quick flows

12.1 Add a car

Register vehicle → first available slot is assigned automatically → Active → QR created. If no slot is available, it remains Unassigned without QR.

12.2 Add a motorcycle

Register motorcycle → Active → no slot → no QR → controlled by registration plate.

12.3 Temporary plate

Active vehicle → Temporary plates → create period → while current, habitual or temporary plate is accepted with the same QR.

12.4 Revoked QR

Vehicle remains Active and keeps its slot → QR no longer authorises access → owner requests renewal → ADMIN approves → new QR is created.

12.5 Suspension / Deactivation

Apply status → operational slot/QR authorisation is removed → activation request is required to restore the vehicle; cars return first as Unassigned.

12.6 Bulk printing

ADMIN filters/selects vehicles → Print QR → motorcycles and records without a valid QR are skipped → 2 cards per A4 using the same credential and size as individual printing.

13. Frequently asked questions and good practices

Situation	What to check
Vehicle is registered but cannot enter	Status, slot and QR. Unassigned/Suspended/Deactivated or Revoked QR do not authorise access.
A car is Unassigned	It has no slot. Assigning one returns it to Active and creates/activates QR authorisation.
QR does not work	Check vehicle status, slot and QR status. If Revoked, request renewal.
There is a temporary plate	The QR does not change; confirm the temporary period is current.
Bulk printing skips a record	Motorcycles, vehicles without a slot or without Active QR are not printed.
Print preview adds date/URL	Disable browser “Headers and footers”.
ADMIN cannot see Control	Switch to CONTROL role if the user has that profile.
Audit is not shown on mobile	Use the desktop version.

13.1 Good practices

- Before applying a sanction, review the incident, follow-ups and actual vehicle involved.
- For QR misuse, distinguish QR owner from observed vehicle.
- Do not use Deactivation as a substitute for Suspension.
- Do not revoke the wrong vehicle QR.
- Keep resolved incidents and requests for traceability.
- Use Data Review before correcting inconsistencies.

> **Support and traceability**
> If behaviour is unexpected, record date, user/role, screen, relevant plate or ID and a screenshot. Audit helps reconstruct what happened.